

KARAN JHA

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Gwalior, Madhya Pradesh 474006, India

PROFESSIONAL SUMMARY

Backend-focused Computer Science Engineering graduate with hands-on experience building secure, scalable REST APIs using Java and Spring Boot. Delivered production-ready projects with JWT authentication, RBAC, and transaction management. Available immediately for a Backend Developer role to solve real engineering problems.

TECHNICAL SKILLS

Languages: Java (Core & Advanced), JavaScript (ES6+), HTML5, CSS3
Backend & Security: Spring Boot, Spring Security, Spring Data JPA, Hibernate, JWT, 3-Tier RBAC, REST APIs
Frontend & Real-Time: React 18, Vite, Tailwind CSS, Recharts, WebSocket's, Socket.IO
Databases & DevOps: PostgreSQL, MySQL, MongoDB, H2, Docker, Git, GitHub Actions (CI/CD), Maven, Postman, Swagger

PROJECTS

Equinox — Finance Management Dashboard 2026

Spring Boot · React · PostgreSQL · JWT · Docker · Render | [GitHub](#)

- **Security & RBAC:** Built stateless JWT authentication and a 3-tier RBAC system (Admin/Analyst/Viewer) using Spring Security @PreAuthorize across 20+ endpoints.
- **Transaction Engine:** Engineered server-side pagination, dynamic JPA Specification filters, soft deletes, and Excel-compatible CSV exports.
- **Database Optimization:** Designed 6 analytics APIs using custom JPQL queries, offloading calculations to the database to eliminate N+1 query overhead.
- **Interactive UI:** Created a responsive React 18 frontend with 6 switchable Recharts views, real-time budget trackers, and mobile-adaptive navigation.
- **DevOps & CI/CD:** Containerized the application via multi-stage Docker builds and deployed on Render/Neon PostgreSQL with automated GitHub workflows.

SkillConnect — Peer-to-Peer Collaboration Platform 2025

React · Socket.IO · WebSocket's · REST APIs | [GitHub](#)

- **Real-Time Collaborative Editor:** Solved fragmented peer-to-peer interview practice by building a live pair-programming workspace using Socket.IO and WebSocket's, synchronizing multi-user code edits and cursor states with sub-100ms latency.
- **Multi-Language Execution Engine:** Integrated a code execution system supporting 5 languages (Java, C, C++, JS, Python), allowing users to run DSA solutions and view instant test-case output feedback directly in the browser.
- **Session State Synchronization:** Built custom WebSocket event handlers to manage live coding room states, handling user connections, disconnections, and payload sync without code loss.
- **RESTful Backend Integration:** Integrated REST endpoints for user authentication, problem catalogue retrieval, and session data persistence, connecting user interface actions to backend database storage.

Book-Pilot Manager — Library Management System 2024

Java · MySQL · Java Swing · NetBeans · JDBC | [GitHub](#)

- **Desktop CRUD Application:** Engineered a desktop management system with 10+ interactive screens, enabling administrators to perform complete CRUD operations on book catalogues and user profiles.
- **Multi-Criteria Search:** Implemented a 5-criteria dynamic search mechanism (filtering by Title, Author, Availability, Book ID, and Student Name) to enable rapid library record lookups.
- **Relational Database Design:** Designed a normalized 4-table MySQL database schema (Users, Students, Books, Issued Transactions) using primary/foreign key relationships to guarantee data consistency.
- **Event-Driven UI Integration:** Integrated Java Swing UI layers with JDBC connectors, using event-driven listeners to instantly synchronize user forms and tables with the database.

EDUCATION

B.Tech in Computer Science Engineering 2022 – 2026
Vikrant Institute of Technology, Gwalior

Higher Secondary (Class XII) 2021 – 2022
Navyug Ch. Hr. Sec. School, Gwalior

CERTIFICATIONS

Oracle AI Foundations Associate — Oracle University, 2025